



Lung Cancer and Covid-19 Diagnosis Techniques using CT Images-A Review

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Abstract:

Cancer sickness occurs by uncontrolled division of erratic cells the lungs. This disease is noteworthy in many parts of the world and keeps rising. Similarly covid-19 disease which occurred due to rapid spread of corona virus all way from Wuhan, China in December 2019 and spread across globally. Covid19 is still continuing in few parts of the world with its variants due to mutation among viruses. Both Lung cancer and covid19 lung diseases damage the lungs due to cellular breakdown and leads to abnormal breathing. These diseases are examined through CAD (Computer Aided Diagnosis) system which is a set of programmed and recognizable proof of instructions by using CT Images (Computed Tomography) to enhance the traditional medical services approach. The lung is checked using CT images which might be impacted by various noises and disturbances. The image is restored using image enhancement techniques to yield a higher order image both in terms quality of brightness and contrast. Then the image is clustered, threshold for extracting the Region of Interest (ROI). Then the image is subjected to Deep Learning algorithms to classify the various phases/stages of the lung diseases (lung cancer and covid19). This review paper gives a gist of the various techniques and algorithms proceeded by researchers and scientists for detection and diagnosis of lung diseases.

Keywords: Cancer, Covid- 19, Enhancement, Restoration, CT images, Deep learning (DL).

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I. Introduction

Lung Diseases are the most infectious and dangerous when not detected and diagnosed earlier. These diseases are caused by abnormal reconstruction of tissues and fibers inside the lung. These may occur due to deposition of extra collagen particles and other extracellular matrix (ECM) components. There are few factors like age, smoking, viral infections, drug exposure and genetic problems which might cause the disease. The person affected cannot breathe at a normal rate due to Excessive Reactive Oxygen Species (ROS) production. Lung diseases occur due to abnormal growth of

infectious cells in lungs connecting the lung parts such as bronchioles, blood vessels etc. These disease have to be diagnosed and detected in an earlier stage which might increase the mortality rate of patients. The prediction of this disease has a few roles like contour, position and volume. Successful detection, staging and treatment phase can be obtained through proper image processing methods. Volumetric Analysis Framework (VAF) gives a clear idea of the processing techniques employed for diagnosis and detection. This method estimates the accuracy of length, thickness, height and type of the detected infectious disease cell.



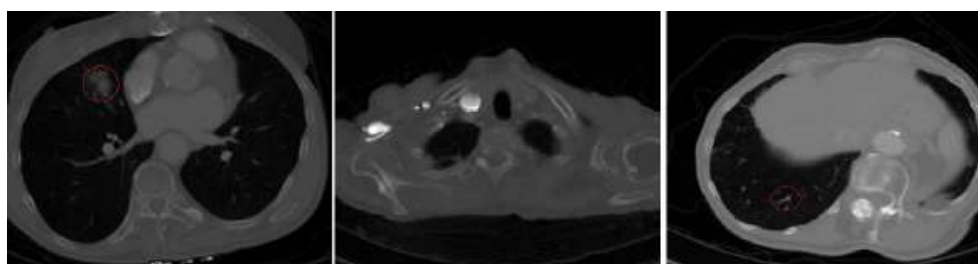


Figure 1: Different angles of Lung CT images

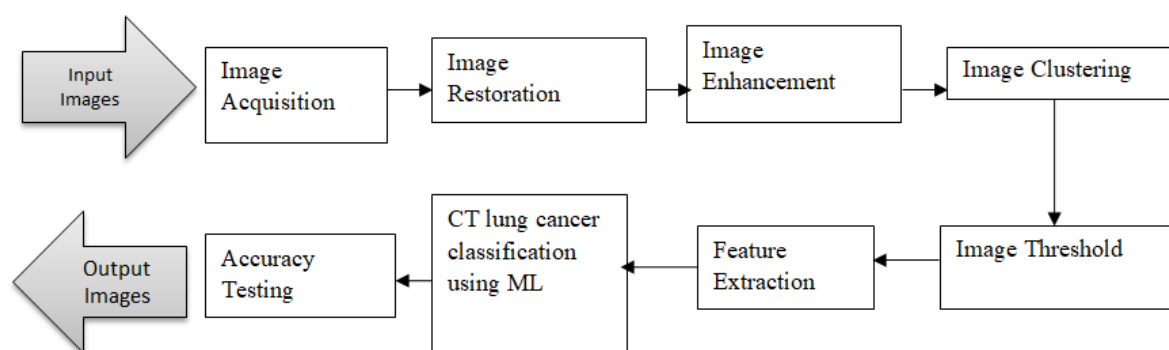


Figure 2: Architecture diagram of identification and classification for lung diseases using CT image

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II. Objectives

Since manual detection is a time consuming process, a fully automated and efficient method, to restore, enhance, segment and threshold the COVID-19 infections in lung CT images have been implemented. Detecting them at an early stage is essential to reduce the mortality risk of the patients. To show the efficiency of the proposed algorithms a few parameters are defined. This automated approach detects the disease from CT images based on deep models for faster examination. A module based on pixels regions for clustering is applied for segmentation. This module increases the ability of the model without information loss.

III. LITERATURE REVIEW

Akila Agnes [1] wanted to bring a full-stoop for manual detection in chest CT and discovered the Convolutional Deep and Wide Network (CDWN). This method segments lung regions from chest CT for further medical diagnosis and obtained 98% accuracy. May Phu Paing[2] identified that manual staging of cancer remains a challenge because of the intensive effort required. Later he proposed CAD (Computer-Aided Diagnosis) method for staging and detecting by using Back-Propagation neural network and

obtained 92.8% for staging and 98.6% for detection.

Xinqi Wang [4] found that early detection of lung disease mainly depends on the reliability of the CT images and tried employing modern Computer Vision and Machine Learning on CT images and came up with 3D-Feature based method which can actually give differences in accuracies using different databases.

Hsin-Jui-Chen[5] identified that automatic Location of lung region in chest X-Ray images is a notable factor when using CAD method. Thus proposed Binarized Enhance Chest X-Ray (BECXR) which could process the CXR using Adaptive pre-processing approach for segmenting the lung regions from CXR and yielded 20.74 % faster training network and 90.6% decrease in storage space

Hashimoto[7] came up with a method called Sinogram Based Dynamic Image Guided Filtering algorithm to enhance the problem of low performance of range distance estimation. This method De-noised the performance in PET(Positron Emission Tomography) and yielded the white matter and gray matter de-noising effects and obtained 92% efficiency

Rodrigues[8] wanted to find a solution for the previous methods that did not provide a proper estimation of Structure and Contour preservation features. He proposed by using



New Association based Smedian Thresholding and Fast BF to remove speckle noise to overcome the issues faced. He brought in the concept of Peak-Signal to Noise Ratio (PSNR) and Mean Structural Similarity Index (MSSIM) and obtained 14.13% increase in PSNR for structural features and 4.96% increase in MSSIM of 0.70% increase in beta values.

Guilio Siracusano [9] improved the sensitivity of CXR with a non-linear post processing tool using PACE method embedded along with BEMD and CLAHE approaches to overcome the risk and cost factors faced when using non-contrast CT images. There was an average increase of 9% for cost investment 2.4% increase in ENT and 2% increase in EME. But this approach was declined since the portability and readability of CXR monitoring of ICU patients was a hectic task.

Jianning Chi [10] proposed the Uniform CNN (convolutional Neural Network) method to handle the de-noising super resolution of CT Images at a time and yielded good precision value of 0.86 and 0.955 in 2 Epochs and recall value of 0.96 and 0.84. The F1 score of 0.91 and 0.89 which brought a major change across the world for the usage of CNN.

Huseyin Polat [12] found that deep learning under machine learning techniques can be used for automatic feature extraction which shall minimize the complicated issues when compared to conventional ML techniques. He employed 3D-AlexNet and 3D Google Net techniques and acquired 85.79% for AlexNet and 87.95% for Google Net method.

Heewon Chung [15] proposed that prediction of the lung image based on segmentation contour in each frame and neighboring upper frame by using Chan-Vese and Bayesian approach. Later he coined his method as Juxtra-Pleural nodal method.

Pietro Nardelli [16] discovered that separating the arteries and veins in CT images will be useful to accurately diagnose the changes in vascular trees and abnormal detection. He proposed the CNN method with Graph cut method random forest classifier and brought 94% of accuracy.

Tao Peng [17] overcame the challenge of handling variations in lung volume, shape and his method automatically detected the lung

boundaries by using Pixel-based Scan Connected Component Labeling Convex Hull Closed Principle Curve (PSCHL-CH-CPC) and obtained an average Disc co-efficient of 96.9%.

Marios Anthimo [18] wanted to detect and recognize Ionized LD which was crucial and challenging even for experienced doctors and proposed the Deep purely CNN for Semantic Segmentation with CAD and acquired CV Balanced accuracy of 81.8%.

Diego Riquelme [19] proposed the State of Art Deep Learning architecture for CAD approaches to detect the false positive reductions. This method was very useful in handling factors like time and detecting malignant nodules in CT. He obtained FP reduction/ 8 CT scans with an accuracy of 87.4% and sensitivity of 91.4%.

Goran Jakimovski [20] came up with a proposal to detect the disease in early by using DNN-Multistage network yielding single decimal values 0.0 which means disease not detected and 1.0 which means disease is detected.

IV. Conclusion

Both Lung cancer and Covid19 are serious illnesses. Since the diseases are spreading in vast amounts doctors and scientific technicians are trying to come up with new methods of detection and diagnosis. This makes the spread take a little slow motion and make people aware of the symptoms through television news, social media etc. so that they could be treated in an earlier stage and thus increases the mortality rate risk. This review paper has discussed about the various image processing techniques from restoration to classification of the disease using published articles and results of several scientific technicians and doctors.

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